Lecture 6
Linear Transformations

6.1 Introduction to Linear Transformations
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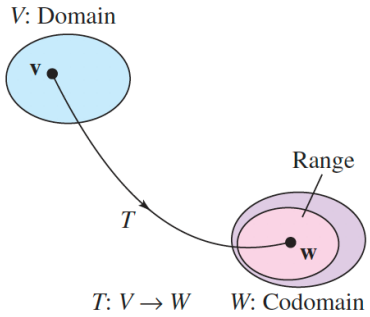
6.1 Introduction to Linear Transformations

- Function T that maps a vector space V into a vector space W:

$$T: V \xrightarrow{\text{mapping}} W, \quad V, W: \text{vector space}$$

V: the domain of T

W: the codomain of T



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- Image of **v** under T:

  If **v** is in V and **w** is in W such that

  $$T(\mathbf{v}) = \mathbf{w}$$

  Then **w** is called the image of **v** under T.

- the range of T:

  The set of all images of vectors in V.

- the preimage of **w**:

  The set of all **v** in V such that $T(\mathbf{v})=\mathbf{w}$.

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- Ex 1: (A function from R^2 into R^2)

  $T: R^2 \to R^2 \qquad \mathbf{v} = (v_1, v_2) \in R^2$

  $T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$

  (*a*) Find the image of **v**=(-1,2). (b) Find the preimage of **w**=(-1,11)

Sol:

(*a*) $\mathbf{v} = (-1, 2)$

$\Rightarrow T(\mathbf{v}) = T(-1, 2) = (-1-2, -1+2(2)) = (-3, 3)$

(*b*) $T(\mathbf{v}) = \mathbf{w} = (-1, 11)$

$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2) = (-1, 11)$

$\Rightarrow v_1 - v_2 = -1$

$v_1 + 2v_2 = 11$

$\Rightarrow v_1 = 3,\ v_2 = 4$ Thus $\{(3, 4)\}$ is the preimage of **w**=(-1, 11).

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- Linear Transformation (L.T.):

V, W – vector spaces
$T: V \to W$, a mapping from V to W, is linear transformation if it satisfies

(1) $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \quad \forall \mathbf{u}, \mathbf{v} \in V$

(2) $T(c\mathbf{u}) = cT(\mathbf{u}), \quad \forall c \in R$

- Notes:

(1) A linear transformation is said to be operation preserving.

$$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \qquad T(c\mathbf{u}) = cT(\mathbf{u})$$

Addition in V | Addition in W | Scalar multiplication in V | Scalar multiplication in W

(2) A linear transformation $T: V \to V$ from a vector space into itself is called a **linear operator**.

- Ex 2: (Verifying a linear transformation T from R^2 into R^2)

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$$

Pf:

$\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$: vector in R^2, c : any real number

(1)Vector addition :

$$\mathbf{u} + \mathbf{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$$

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T(u_1 + v_1, u_2 + v_2) \\ &= ((u_1 + v_1) - (u_2 + v_2), (u_1 + v_1) + 2(u_2 + v_2)) \\ &= ((u_1 - u_2) + (v_1 - v_2), (u_1 + 2u_2) + (v_1 + 2v_2)) \\ &= (u_1 - u_2, u_1 + 2u_2) + (v_1 - v_2, v_1 + 2v_2) \\ &= T(\mathbf{u}) + T(\mathbf{v}) \end{aligned}$$

(2) Scalar multiplication

$$c\mathbf{u} = c(u_1, u_2) = (cu_1, cu_2)$$

$$\begin{aligned} T(c\mathbf{u}) &= T(cu_1, cu_2) = (cu_1 - cu_2, cu_1 + 2cu_2) \\ &= c(u_1 - u_2, u_1 + 2u_2) \\ &= cT(\mathbf{u}) \end{aligned}$$

Therefore, T is a linear transformation.

- Ex 3: (Functions that are not linear transformations)

$(a) f(x) = \sin x$

$\sin(x_1 + x_2) \neq \sin(x_1) + \sin(x_2)$

$\sin(\frac{\pi}{2} + \frac{\pi}{3}) \neq \sin(\frac{\pi}{2}) + \sin(\frac{\pi}{3})$

$\Leftarrow f(x) = \sin x$ is not linear transformation

$(b) f(x) = x^2$

$(x_1 + x_2)^2 \neq x_1^2 + x_2^2$

$(1+2)^2 \neq 1^2 + 2^2$

$\Leftarrow f(x) = x^2$ is not linear transformation

$(c) f(x) = x + 1$

$f(x_1 + x_2) = x_1 + x_2 + 1$

$f(x_1) + f(x_2) = (x_1 + 1) + (x_2 + 1) = x_1 + x_2 + 2$

$f(x_1 + x_2) \neq f(x_1) + f(x_2)$

$\Leftarrow f(x) = x + 1$ is not linear transformation

- Notes: Two uses of the term "linear".

(1) $f(x) = x + 1$ is called a linear function because its graph is a line.

(2) $f(x) = x + 1$ is not a linear transformation from a vector space R into R because it preserves neither vector addition nor scalar multiplication.

- Zero transformation:

$T: V \to W \qquad T(\mathbf{v}) = 0, \ \forall \mathbf{v} \in V$

- Identity transformation:

$T: V \to V \qquad T(\mathbf{v}) = \mathbf{v}, \ \forall \mathbf{v} \in V$

- Thm 6.1: (Properties of linear transformations)

$T: V \to W, \quad \mathbf{u}, \mathbf{v} \in V$

(1) $T(\mathbf{0}) = \mathbf{0}$

(2) $T(-\mathbf{v}) = -T(\mathbf{v})$

(3) $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$

(4) If $\mathbf{v} = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$

Then $T(\mathbf{v}) = T(c_1 v_1 + c_2 v_2 + \cdots + c_n v_n)$

$= c_1 T(v_1) + c_2 T(v_2) + \cdots + c_n T(v_n)$

- Ex 4: (Linear transformations and bases)

Let $T: R^3 \to R^3$ be a linear transformation such that

$T(1,0,0) = (2,-1,4)$

$T(0,1,0) = (1,5,-2)$

$T(0,0,1) = (0,3,1)$

Recall:
$\mathbf{e_1}=(1,0,0)$; $\mathbf{e_2}=(0,1,0)$; $\mathbf{e_3}=(0,0,1)$ is the standard basis of R^3

Find $T(2, 3, -2)$.

Sol:

$(2,3,-2) = 2(1,0,0) + 3(0,1,0) - 2(0,0,1)$

$T(2,3,-2) = 2T(1,0,0) + 3T(0,1,0) - 2T(0,0,1)$ $\Leftarrow$ because (T is a L.T.)

$= 2(2,-1,4) + 3(1,5,-2) - 2T(0,3,1)$

$= (7,7,0)$

- Ex 5: (A linear transformation defined by a matrix)

The function $T: R^2 \to R^3$ is defined as $T(\mathbf{v}) = A\mathbf{v} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

(a) Find $T(\mathbf{v})$, where $\mathbf{v} = (2,-1)$

(b) Show that T is a linear transformation form R^2 into R^3

Sol: $(a)\mathbf{v} = (2,-1)$ (R^2 vector, R^3 vector)

$$T(\mathbf{v}) = A\mathbf{v} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix}$$

$$\therefore T(2,-1) = (6,3,0)$$

$(b)\ T(\mathbf{u}+\mathbf{v}) = A(\mathbf{u}+\mathbf{v}) = A\mathbf{u} + A\mathbf{v} = T(\mathbf{u}) + T(\mathbf{v})$ (vector addition)

$T(c\mathbf{u}) = A(c\mathbf{u}) = c(A\mathbf{u}) = cT(\mathbf{u})$ (scalar multiplication)

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- Thm 6.2: (The linear transformation given by a matrix)

Let A be an $m \times n$ matrix. The function T defined by

$$T(\mathbf{v}) = A\mathbf{v}$$

is a linear transformation $T: R^n \to R^m$.

- Note: (R^n vector, R^m vector)

$$A\mathbf{v} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n \\ a_{21}v_1 + a_{22}v_2 + \cdots + a_{2n}v_n \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \cdots + a_{mn}v_n \end{bmatrix}$$

$$T: R^n \to R^m$$

$$T(\mathbf{v}) = A\mathbf{v}$$

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- Ex 7: (Rotation in the plane)

Show that the L.T. $T: R^2 \to R^2$ given by the matrix

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

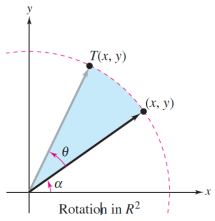
has the property that it rotates every vector in R^2 counterclockwise about the origin through the angle θ.

Sol:

$v = (x, y) = (r\cos\alpha, r\sin\alpha)$ (polar coordinates)

r : the length of v

α : the angle from the positive x-axis counterclockwise to the vector v



$$T(\mathbf{v}) = A\mathbf{v} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} r\cos\alpha \\ r\sin\alpha \end{bmatrix}$$

$$= \begin{bmatrix} r\cos\theta\cos\alpha - r\sin\theta\sin\alpha \\ r\sin\theta\cos\alpha + r\cos\theta\sin\alpha \end{bmatrix}$$

$$= \begin{bmatrix} r\cos(\theta+\alpha) \\ r\sin(\theta+\alpha) \end{bmatrix}$$

r : the length of $T(\mathbf{v})$

$\theta+\alpha$: the angle from the positive x-axis counterclockwise to the vector $T(\mathbf{v})$

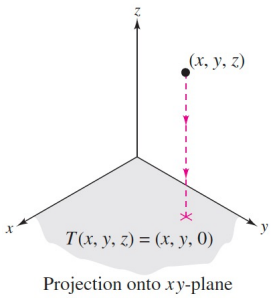
Thus, $T(\mathbf{v})$ is the vector that results from rotating the vector $\mathbf{v}$ counterclockwise through the angle θ.

- Ex 8: (A projection in R^3)

The linear transformation $T: R^3 \to R^3$ is given by

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is called a projection in R^3.



Projection onto xy-plane

- Ex 9: (A linear transformation from $M_{m\times n}$ into $M_{n\times m}$)

$T(A) = A^T \quad (T: M_{m\times n} \to M_{n\times m})$

Show that T is a linear transformation.

Sol:

$A, B \in M_{m\times n}$

$T(A+B) = (A+B)^T = A^T + B^T = T(A) + T(B)$

$T(cA) = (cA)^T = cA^T = cT(A)$

Therefore, T is a linear transformation from $M_{m\times n}$ into $M_{n\times m}$.

6.2 The Kernel and Range of a Linear Transformation

- Kernel of a linear transformation T:

  Let $T: V \to W$ be a linear transformation

  Then the set of all vectors $\mathbf{v}$ in V that satisfy $T(\mathbf{v}) = 0$ is called the kernel of T and is denoted by $\ker(T)$.

  $$\ker(T) = \{\mathbf{v} \mid T(\mathbf{v}) = 0, \forall \mathbf{v} \in V\}$$

- Ex 1: (Finding the kernel of a linear transformation)

  $T(A) = A^T \quad (T: M_{3\times 2} \to M_{2\times 3})$

  Sol:

  $$\ker(T) = \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \right\}$$

- Ex 2: (The kernel of the zero and identity transformations)

  (a) $T(\mathbf{v})=\mathbf{0}$ (the zero transformation $T: V \to W$)

  $$\ker(T) = V$$

  (b) $T(\mathbf{v})=\mathbf{v}$ (the identity transformation $T: V \to V$)

  $$\ker(T) = \{\mathbf{0}\}$$

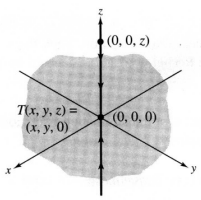
- Ex 3: (Finding the kernel of a linear transformation)

  $T(x, y, z) = (x, y, 0) \qquad (T: R^3 \to R^3)$

  $\ker(T) = ?$

  Sol:

  $\ker(T) = \{(0,0,z) \mid z \text{ is a real number}\}$



The kernel of T is the set of all vectors on the z-axis.

- Ex 5: (Finding the kernel of a linear transformation)

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad (T: R^3 \to R^2)$$

$\ker(T) = ?$

Sol:

$$\ker(T) = \{(x_1, x_2, x_3) \mid T(x_1, x_2, x_3) = (0,0), x = (x_1, x_2, x_3) \in R^3\}$$

$$T(x_1, x_2, x_3) = (0,0)$$

$$\begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \text{or} \begin{cases} x_1 - x_2 - 2x_3 = 0 \\ -x_1 + 2x_2 + 3x_3 = 0 \end{cases}$$

$$\begin{bmatrix} 1 & -1 & -2 & 0 \\ -1 & 2 & 3 & 0 \end{bmatrix} \xrightarrow{G.J.E} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \ker(T) = \{t(1,-1,1) \mid t \text{ is a real number}\}$$
$$= \text{span}\{(1,-1,1)\}$$

- Thm 6.3: (The kernel is a subspace of V)

  The kernel of a linear transformation $T: V \to W$ is a subspace of the domain V.

  Pf: $\because T(0) = 0$ (Theorem 4.1)

  $\therefore \ker(T)$ is a nonempty subset of V

  Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the kernel of T. then

  $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) = 0 + 0 = 0 \quad \Rightarrow \mathbf{u}+\mathbf{v} \in \ker(T)$

  $T(c\mathbf{u}) = cT(\mathbf{u}) = c0 = 0 \quad \Rightarrow c\mathbf{u} \in \ker(T)$

  Thus, $\ker(T)$ is a subspace of V.

- Note:

  The kernel of T is sometimes called the **nullspace** of T.

- Ex 6: (Finding a basis for the kernel)

  Let $T: R^5 \to R^4$ be defined by $T(\mathbf{x}) = A\mathbf{x}$, where $\mathbf{x}$ is in R^5 and

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 \\ -1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 8 \end{bmatrix}$$

  Find a basis for $\ker(T)$ as a subspace of R^5.

Sol:

$$[A \mid 0] =$$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & -1 & 0 \\ 2 & 1 & 3 & 1 & 0 & 0 \\ -1 & 0 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 8 & 0 \end{bmatrix} \xrightarrow{G.J.E} \begin{bmatrix} 1 & 0 & 2 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

s t

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2s+t \\ s+2t \\ s \\ -4t \\ t \end{bmatrix} = s\begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} 1 \\ 2 \\ 0 \\ -4 \\ 1 \end{bmatrix}$$

$B = \{(-2,1,1,0,0),(1,2,0,-4,1)\}$: one basis for the kernel of T

- Corollary to Thm 6.3:

Let $T: R^n \to R^m$ be the L.T given by $T(\mathbf{x}) = A\mathbf{x}$

Then the kernel of T is equal to the solution space of $A\mathbf{x} = 0$

$T(\mathbf{x}) = A\mathbf{x}$ (a linear transformation $T: R^n \to R^m$)

$\Rightarrow Ker(T) = NS(A) = \{\mathbf{x} \mid A\mathbf{x} = 0, \forall \mathbf{x} \in R^m\}$ (subspace of R^m)

- Range of a linear transformation T:

Let $T: V \to W$ is a linear transformation. Then the set of all vectors $w \in W$ that are images of vectors in V is called the range of T and is denoted by $range(T)$.

$$range(T) = \{w \in W \text{ such that } w = T(v) \text{ for some } v \in V\}$$

Or, equivalently $range(T) = \{T(\mathbf{v}) \mid \forall \mathbf{v} \in V\}$

- Thm 6.4: (The range of T is a subspace of W)

The range of a linear transformation $T: V \to W$ is a subspace of W.

Pf:

$\because T(0) = 0$ (Thm.6.1)

$\therefore range(T)$ is a nonempty subset of W

Let $T(\mathbf{u})$ and $T(\mathbf{v})$ be vector in the range of T

$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \in range(T)$ $(\mathbf{u} \in V, \mathbf{v} \in V \Rightarrow \mathbf{u}+\mathbf{v} \in V)$

$T(c\mathbf{u}) = cT(\mathbf{u}) \in range(T)$ $(\mathbf{u} \in V \Rightarrow c\mathbf{u} \in V)$

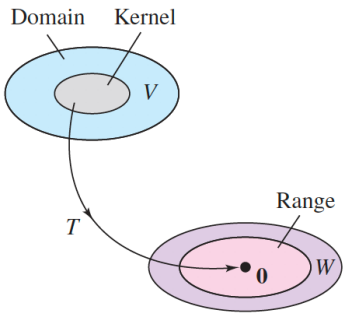
Therefore, $range(T)$ is W subspace.

- Notes:

$T: V \to W$ is a L.T.

(1) $Ker(T)$ is subspace of V

(2) $range(T)$ is subspace of W



- Corollary to Thm 6.4:

Let $T: R^n \to R^m$ be the L.T. given by $T(\mathbf{x}) = A\mathbf{x}$

Then the range of T is equal to the column space of A

$\Rightarrow range(T) = CS(A)$

- Ex 7: (Finding a basis for the range of a linear transformation)

Let $T: R^5 \to R^4$ be defined by $T(\mathbf{x}) = A\mathbf{x}$, where $\mathbf{x}$ is R^5 and

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 \\ -1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 8 \end{bmatrix}$$

Find a basis for the range of T.

Sol:

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 \\ -1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 8 \end{bmatrix} \xrightarrow{G.J.E} \begin{bmatrix} \textcircled{1} & 0 & 2 & 0 & -1 \\ 0 & \textcircled{1} & -1 & 0 & 2 \\ 0 & 0 & 0 & \textcircled{1} & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = B$$

$$\begin{matrix} c_1 & c_2 & c_3 & c_4 & c_5 \end{matrix} \qquad \begin{matrix} w_1 & w_2 & w_3 & w_4 & w_5 \end{matrix}$$

$\Rightarrow \{w_1, w_2, w_4\}$ is a basis for $CS(B)$

$\{c_1, c_2, c_4\}$ is a basis for $CS(A)$

$\Rightarrow \{(1, 2, -1, 0), (2, 1, 0, 0), (1, 1, 0, 2)\}$ is a basis for the range of T

- Rank of a linear transformation $T: V \to W$:

  $rank(T) = $ the dimension of the range of T

- Nullity of a linear transformation $T: V \to W$:

  $nullity(T) = $ the dimension of the kernel of T

- Note:

  Let $T: R^n \to R^m$ be the L.T. given by $T(\mathbf{x}) = A\mathbf{x}$, then

  $rank(T) = rank(A)$

  $nullity(T) = nullity(A)$

- Thm 6.5: (Sum of rank and nullity)

  Let $T: V \to W$ be a L.T. form an n - dimensional vector space V into a vector space W. Then

  $rank(T) + nullity(T) = n$

  $\dim(\text{range of } T) + \dim(\text{kernel of } T) = \dim(\text{domain of } T)$

Pf:

Let T is represented by an $m \times n$ *matrix* A

Assume $rank(A) = r$

$(1) rank(T) = \dim(\text{range of } T) = \dim(\text{column space of } A)$

$= rank(A) = r$

$(2) nullity(T) = \dim(\text{kernel of } T) = \dim(\text{solution space of } A)$

$= n - r$

$\Rightarrow rank(T) + nullity(T) = r + (n - r) = n$

- Ex 8: (Finding the rank and nullity of a linear transformation)

Find the rank and nullity of the L.T. $T: R^3 \to R^3$ define by

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Sol:

$rank(T) = rank(A) = 2$

$nullity(T) = \dim(\text{domain of } T) - rank(T) = 3 - 2 = 1$

- Ex 9: (Finding the rank and nullity of a linear transformation)

Let $T: R^5 \to R^7$ be a linear transformation.

(a) Find the dimension of the kernel of T if the dimension of the range is 2

(b) Find the rank of T if the nullity of T is 4

(c) Find the rank of T if $Ker(T) = \{0\}$

Sol:

$(a)\ \dim(\text{domain of } T) = 5$

$\dim(\text{kernel of } T) = n - \dim(\text{range of } T) = 5 - 2 = 3$

$(b) rank(T) = n - nullity(T) = 5 - 4 = 1$

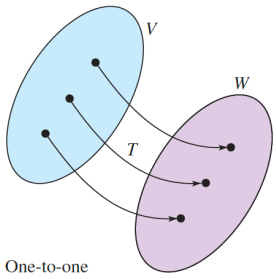
$(c) rank(T) = n - nullity(T) = 5 - 0 = 5$

- One-to-one:

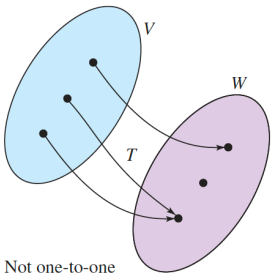
  A function $T : V \to W$ is called one - to - one if the preimage of every w in the range consists of a single vector.

  In other words,

  T is one - to - one iff for all u and v inV, $T(\mathbf{u}) = T(\mathbf{v})$ implies that $\mathbf{u} = \mathbf{v}$.



One-to-one



Not one-to-one

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- Onto:

  A function $T : V \to W$ is said to be onto if every element in **w** has a preimage in V

  (T is onto W when W is equal to the range of T.)

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- Thm 6.6: (One-to-one linear transformation)

Let $T: V \to W$ be a L.T.

Then T is 1-1 iff $Ker(T) = \{0\}$

Pf:

Suppose T is 1-1

Then $T(v) = 0$ can have only one solution : $v = 0$

i.e. $Ker(T) = \{0\}$

Suppose $Ker(T) = \{0\}$ and $T(u) = T(v)$

$T(u - v) = T(u) - T(v) = 0$

(T is a L.T.)

$\because u - v \in Ker(T) \Rightarrow u - v = 0$

$\Rightarrow T$ is 1-1

- Ex 10: (One-to-one and not one-to-one linear transformation)

(a) The L.T. $T: M_{m\times n} \to M_{n\times m}$ given by $T(A) = A^T$ is one - to - one.

Because its kernel consists of only the $m \times n$ zero matrix.

(b) The zero transformation $T: R^3 \to R^3$ is not one - to - one.

Because its kernel is all of R^3.

- Thm 6.7: (Onto linear transformation)

  Let $T: V \to W$ be a L.T., where W is finite dimensional.

  Then T is onto iff the rank of T is equal to the dimension of W.

- Thm 6.8: (One-to-one and onto linear transformation)

  Let $T: V \to W$ be a L.T. with vector space V and W both of dimension n. Then T is one - to - one if and only if it is onto.

Pf:

If T is one - to - one, then $Ker(T) = \{0\}$ and $\dim(Ker(T)) = 0$

$\dim(range(T)) = n - \dim(Ker(T)) = n = \dim(W)$

Consequently, T is onto.

If T is onto, then $\dim(\text{range of } T) = \dim(W) = n$

$\dim(Ker(T)) = n - \dim(\text{range of } T) = n - n = 0$

Therefore, T is one - to - one.

- Ex 11:

The L.T. $T: R^n \to R^m$ is given by $T(\mathbf{x}) = A\mathbf{x}$, Find the nullity and rank of T and determine whether T is one - to - one, onto, or neither.

$$(a)A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \qquad (b)A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$(c)A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \end{bmatrix} \qquad (d)A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Sol:

$T: R^n \to R^m$	dim(domain of T)	rank(T)	nullity(T)	1-1	onto
$(a)T: R^3 \to R^3$	3	3	0	Yes	Yes
$(b)T: R^2 \to R^3$	2	2	0	Yes	No
$(c)T: R^3 \to R^2$	3	2	1	No	Yes
$(d)T: R^3 \to R^3$	3	2	1	No	No

- Isomorphism:

  A linear transformation $T: V \to W$ that is one to one and onto is called an isomorphism. Moreover, if V and W are vector spaces such that there exists an isomorphism from V to W, then V and W are said to be isomorphic to each other.

- Thm 6.9: (Isomorphic spaces and dimension)

  Two finite-dimensional vector space V and W are isomorphic if and only if they are of the same dimension.

  Pf:

  Assume that V is isomorphic to W, where V has dimension n.

$\Rightarrow$ There exists a L.T. $T: V \to W$ that is one to one and onto.

$\because T$ is one - to - one

$\Rightarrow \dim(Ker(T)) = 0$

$\Rightarrow \dim(\text{range of } T) = \dim(\text{domain of } T) - \dim(Ker(T)) = n - 0 = n$

$\because T$ is onto.

$\Rightarrow \dim(\text{range of } T) = \dim(W) = n$

Thus $\dim(V) = \dim(W) = n$

Assume that V and W both have dimension n.

Let $\{v_1, v_2, \cdots, v_n\}$ be a basis of V, and

let $\{w_1, w_2, \cdots, w_n\}$ be a basis of W.

Then an arbitrary vector in V can be represented as

$\mathbf{v} = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$

and you can define a L.T. $T: V \to W$ as follows.

$T(\mathbf{v}) = c_1 w_1 + c_2 w_2 + \cdots + c_n w_n$

It can be shown that this L.T. is both 1-1 and onto.

Thus V and W are isomorphic.

Home work #1: Verify if following linear transformations are one-to-one, onto or isomorphism

a) $f: R^2 \longrightarrow R^1$ defined by $f(x_1, x_2) = x_1$ - *the projection*

b) $f: R^1 \longrightarrow R^2$ defined by $f(x_1) = (x_1, x_1)$ – *the diagonal*

c) $f: R^2 \longrightarrow R^2$ defined by $f(x_1, x_2) = (x_2, x_1)$ – *the reflection*

- Ex 12: (Isomorphic vector spaces)

The following vector spaces are isomorphic to each other.

$(a) R^4 = 4$-space

$(b) M_{4\times1} =$ space of all 4×1 matrices

$(c) M_{2\times2} =$ space of all 2×2 matrices

$(d) P_3(x) =$ space of all polynomials of degree 3 or less

$(e) V = \{(x_1, x_2, x_3, x_4, 0), x_i \text{ is a real number}\}$ (subspace of R^5)

6.3 Matrices for Linear Transformations

- Two representations of the linear transformation $T:R^3 \to R^3$:

$$(1) T(x_1, x_2, x_3) = (2x_1 + x_2 - x_3, -x_1 + 3x_2 - 2x_3, 3x_2 + 4x_3)$$

$$(2) T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 2 & 1 & -1 \\ -1 & 3 & -2 \\ 0 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- Three reasons for matrix representation of a linear transformation:
  - It is simpler to write.
  - It is simpler to read.
  - It is more easily adapted for computer use.

- Thm 6.10: (**Standard** matrix for a linear transformation)

Let $T : R^n \to R^m$ be a linear transformation such that

$$T(e_1) = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \quad \cdots, \quad T(e_n) = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix},$$

Where $\{e_1, e_2, \ldots, e_n\}$ is the standard basis of R^n.

Then the $m \times n$ matrix whose n columns correspond to $T(e_i)$

$$A = [T(e_1) \mid T(e_2) \mid \cdots \mid T(e_n)] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

is such that $T(\mathbf{v}) = A\mathbf{v}$ for every $\mathbf{v}$ in R^n.

A is called the standard matrix for T.

Pf:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1 e_1 + v_2 e_2 + \cdots + v_n e_n$$

$$\begin{aligned} T \text{ is a L.T.} \Rightarrow T(\boldsymbol{v}) &= T(v_1 e_1 + v_2 e_2 + \cdots + v_n e_n) \\ &= T(v_1 e_1) + T(v_2 e_2) + \cdots + T(v_n e_n) \\ &= v_1 T(e_1) + v_2 T(e_2) + \cdots + v_n T(e_n) \end{aligned}$$

$$A\mathbf{v} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n \\ a_{21}v_1 + a_{22}v_2 + \cdots + a_{2n}v_n \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \cdots + a_{mn}v_n \end{bmatrix}$$

$$= v_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + v_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + v_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

$$= v_1 T(e_1) + v_2 T(e_2) + \cdots + v_n T(e_n)$$

Therefore, $T(\mathbf{v}) = A\mathbf{v}$ for each $\mathbf{v}$ in R^n

- Ex 1: (Finding the standard matrix of a linear transformation)

Find the standard matrix for the L.T. $T: R^3 \to R^2$ define by

$$T(x, y, z) = (x - 2y, 2x + y)$$

Sol:

Vector Notation	Matrix Notation
$T(e_1) = T(1, 0, 0) = (1, 2)$	$T(e_1) = T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
$T(e_2) = T(0, 1, 0) = (-2, 1)$	$T(e_2) = T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$
$T(e_3) = T(0, 0, 1) = (0, 0)$	$T(e_3) = T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

USTH

$$A = [T(e_1) \mid T(e_2) \mid T(e_3)]$$

$$= \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \end{bmatrix}$$

- Check:

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x - 2y \\ 2x + y \end{bmatrix}$$

i.e. $T(x, y, z) = (x - 2y, 2x + y)$

- Note:

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \end{bmatrix} \begin{matrix} \leftarrow 1x - 2y + 0z \\ \leftarrow 2x + 1y + 0z \end{matrix}$$

USTH

- Ex 2: (Finding the standard matrix of a linear transformation)
  The linear transformation $T: R^2 \to R^2$ is given by projecting each point in R^2 onto the x - axis. Find the standard matrix for T.

  Sol:

$$T(x, y) = (x, 0)$$

$$A = [T(e_1) \mid T(e_2)] = [T(1, 0) \mid T(0, 1)] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

- Notes:

(1) The standard matrix for the zero transformation from R^n into R^m is the mxn zero matrix.

(2) The standard matrix for the identity transformation from R^n into R^n is the nxn identity matrix I_n

Home work #2: Find the standard matrix of following linear transformations

a) $f: R^2 \longrightarrow R^1$ defined by $f(x_1, x_2) = x_1$

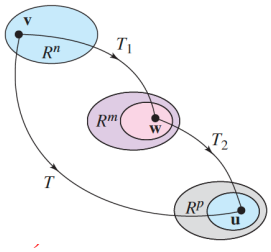
b) $f: R^1 \longrightarrow R^2$ defined by $f(x_1) = (x_1, x_1)$

c) $f: R^2 \longrightarrow R^2$ defined by $f(x_1, x_2) = (x_2, x_1)$

- Composition of $T_1: R^n \to R^m$ with $T_2: R^m \to R^p$:

$$T(\mathbf{v}) = T_2(T_1(\mathbf{v})), \quad \mathbf{v} \in R^n$$

$$T = T_2 \circ T_1, \quad \text{domain of } T = \text{domain of } T_1$$



- Thm 6.11: (Composition of linear transformati

Let $T_1 : R^n \to R^m$ and $T_2 : R^m \to R^p$ be L.T.
with standard matrices A_1 and A_2 , then

(1)The composition $T : R^n \to R^p$, defined by $T(\mathbf{v}) = T_2(T_1(\mathbf{v}))$, is a L.T.

(2) The standard matrix A for T is given by the matrix product $A = A_2 A_1$

Pf:

(1)(T is a L.T.)

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in R^n and let c be any scalar then

$$T(\mathbf{u}+\mathbf{v}) = T_2(T_1(\mathbf{u}+\mathbf{v})) = T_2(T_1(\mathbf{u}) + T_1(\mathbf{v}))$$
$$= T_2(T_1(\mathbf{u})) + T_2(T_1(\mathbf{v})) = T(\mathbf{u}) + T(\mathbf{v})$$

$$T(c\mathbf{v}) = T_2(T_1(c\mathbf{v})) = T_2(cT_1(\mathbf{v})) = cT_2(T_1(\mathbf{v})) = cT(\mathbf{v})$$

(2)($A_2 A_1$ is the standard matrix for T)

$$T(\mathbf{v}) = T_2(T_1(\mathbf{v})) = T_2(A_1\mathbf{v}) = A_2A_1\mathbf{v} = (A_2A_1)\mathbf{v}$$

- Note:

$$T_1 \circ T_2 \neq T_2 \circ T_1$$

- Ex 3: (The standard matrix of a composition)

Let T_1 and T_2 be L.T. from R^3 into R^3 s.t.

$$T_1(x,y,z) = (2x+y, 0, x+z)$$
$$T_2(x,y,z) = (x-y, z, y)$$

Find the standard matrices for the compositions

$T = T_2 \circ T_1$ and $T' = T_1 \circ T_2$,

Sol:

$$A_1 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \text{ (standard matrix for } T_1)$$

$$A_2 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ (standard matrix for } T_2)$$

The standard matrix for $T = T_2 \circ T_1$

$$A = A_2 A_1 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The standard matrix for $T' = T_1 \circ T_2$

$$A' = A_1 A_2 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -2 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

- Inverse linear transformation:

If $T_1 : R^n \to R^n$ and $T_2 : R^n \to R^n$ are L.T. s.t. for every $\mathbf{v}$ in R^n

$$T_2(T_1(\mathbf{v})) = \mathbf{v} \quad \text{and} \quad T_1(T_2(\mathbf{v})) = \mathbf{v}$$

Then T_2 is called the inverse of T_1 and T_1 is said to be invertible

- Note:

If the transformation T is invertible, then the inverse is unique and denoted by T^{-1}.

- Thm 6.12: (Existence of an inverse transformation)

Let $T : R^n \to R^n$ be a L.T. with standard matrix A,
Then the following condition are equivalent.

(1) T is invertible.

(2) T is an isomorphism.

(3) A is invertible.

- Note:

If T is invertible with standard matrix A, then the standard matrix for T^{-1} is A^{-1}.

- Ex 4: (Finding the inverse of a linear transformation)

The L.T. $T: R^3 \to R^3$ is defined by

$$T(x_1, x_2, x_3) = (2x_1 + 3x_2 + x_3, 3x_1 + 3x_2 + x_3, 2x_1 + 4x_2 + x_3)$$

Show that T is invertible, and find its inverse.

Sol:

The standard matrix for T

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 3 & 1 \\ 2 & 4 & 1 \end{bmatrix} \quad \begin{matrix} \leftarrow 2x_1 + 3x_2 + x_3 \\ \leftarrow 3x_1 + 3x_2 + x_3 \\ \leftarrow 2x_1 + 4x_2 + x_3 \end{matrix}$$

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 2 & 3 & 1 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 1 & 0 \\ 2 & 4 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{G.J.E} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & 6 & -2 & -3 \end{array}\right] = [I \mid A^{-1}]$$

Therefore T is invertible and the standard matrix for T^{-1} is A^{-1}

$$A^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix}$$

$$T^{-1}(\mathbf{v}) = A^{-1}\mathbf{v} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_1 + x_2 \\ -x_1 + x_3 \\ 6x_1 - 2x_2 - 3x_3 \end{bmatrix}$$

In other words,

$$T^{-1}(x_1, x_2, x_3) = (-x_1 + x_2, -x_1 + x_3, 6x_1 - 2x_2 - 3x_3)$$

Matrix of a LT in general case

- Transformation matrix for nonstandard bases:

Let V and W be finite-dimensional vector spaces with basis B and B', respectively, where $B = \{v_1, v_2, \cdots, v_n\}$

If $T : V \to W$ is a L.T. s.t.

$$[T(v_1)]_{B'} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, \quad [T(v_2)]_{B'} = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \cdots, \quad [T(v_n)]_{B'} = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

then the $m \times n$ matrix whose n columns correspond to $[T(v_i)]_{B'}$

$$A = \left[\; T(v_1) \; \middle| \; T(v_2) \; \middle| \; \cdots \; \middle| \; T(v_n) \; \right] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

is such that $[T(\mathbf{v})]_{B'} = A[\mathbf{v}]_B$ for every $\mathbf{v}$ in V.

A is called the matrix of the transformation T relative to bases B and B'.

- Ex 5: (Finding a matrix relative to nonstandard bases)

Let $T: R^2 \to R^2$ be a L.T. defined by

$$T(x_1, x_2) = (x_1 + x_2, 2x_1 - x_2)$$

Find the matrix of T relative to the basis

$B = \{(1, 2), (-1, 1)\}$ and $B' = \{(1, 0), (0, 1)\}$

Sol:

$$T(1, 2) = (3, 0) = 3(1, 0) + 0(0, 1)$$

$$T(-1, 1) = (0, -3) = 0(1, 0) - 3(0, 1)$$

$$[T(1, 2)]_{B'} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad [T(-1, 1)]_{B'} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$$

the matrix for T relative to B and B'

$$A = [[T(1, 2)]_{B'} \quad [T(1, 2)]_{B'}] = \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix}$$

- Ex 6:

For the L.T. $T: R^2 \to R^2$ given in Example 5, use the matrix A to find $T(\mathbf{v})$, where $\mathbf{v} = (2, 1)$

Sol:

$$\mathbf{v} = (2, 1) = 1(1, 2) - 1(-1, 1) \qquad B = \{(1, 2), (-1, 1)\}$$

$$\Rightarrow [\mathbf{v}]_B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow [T(\mathbf{v})]_{B'} = A[\mathbf{v}]_B = \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\Rightarrow T(\mathbf{v}) = 3(1, 0) + 3(0, 1) = (3, 3) \qquad B' = \{(1, 0), (0, 1)\}$$

- Check:

$$T(2, 1) = (2 + 1, 2(2) - 1) = (3, 3)$$

- Summary: the matrix of T relative to the bases B and B':

$T: V \to W$ (a L.T.)

$B = \{v_1, v_2, \cdots, v_n\}$ (a basis for V)

$B' = \{w_1, w_2, \cdots, w_m\}$ (a basis for W)

Thus, the matrix of T relative to the bases B *and* B' is

$$A = \left[[T(v_1)]_{B'}, [T(v_2)]_{B'}, \cdots, [T(v_n)]_{B'}\right] \in M_{m\times n}$$

and we have $[T(\mathbf{v})]_{B'} = A[\mathbf{v}]_B$ for every $\mathbf{v}$ in V.

- Notes:

(1)In the special case where $V = W$ and $B = B'$,

the matrix A is called the matrix of T relative to the basis B

(2)$T: V \to V$: the identity transformation

$B = \{v_1, v_2, \cdots, v_n\}$: a basis for V

$\Rightarrow$ the matrix of T relative to the basis B

$$A = \left[[T(v_1)]_B, [T(v_2)]_B, \cdots, [T(v_n)]_B\right] = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I_n$$

6.4 Transition Matrices and Similarity

Suppose that $T: V \longrightarrow V$ is a LT. We have $\mathbf{v} \longmapsto T(\mathbf{v})$ for any $\mathbf{v} \in V$

Suppose that $B = \{v_1, v_2, \ldots, v_n\}$ is a basis for V

Suppose that $B' = \{w_1, w_2, \ldots, w_n\}$ is another basis for V

$A = [[T(v_1)]_B, [T(v_2)]_B, \cdots, [T(v_n)]_B]$ (matrix of T relative to B)

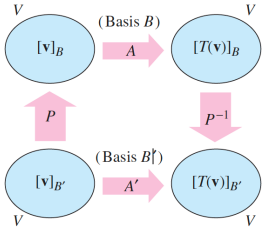
$A' = [[T(w_1)]_{B'}, [T(w_2)]_{B'}, \cdots, [T(w_n)]_{B'}]$ (matrix of T relative to B')

$P = [[w_1]_B, [w_2]_B, \cdots, [w_n]_B]$ (transition matrix from B' to B)

$P^{-1} = [[v_1]_{B'}, [v_2]_{B'}, \cdots, [v_n]_{B'}]$ (transition matrix from B to B')

$[\mathbf{v}]_B = P[\mathbf{v}]_{B'}$ $\quad$ $[\mathbf{v}]_{B'} = P^{-1}[\mathbf{v}]_B$

$[T(\mathbf{v})]_B = A[\mathbf{v}]_B$ $\quad$ $[T(\mathbf{v})]_{B'} = A'[\mathbf{v}]_{B'}$



- Two ways to get from $[\mathbf{v}]_{B'}$ to $[T(\mathbf{v})]_{B'}$:

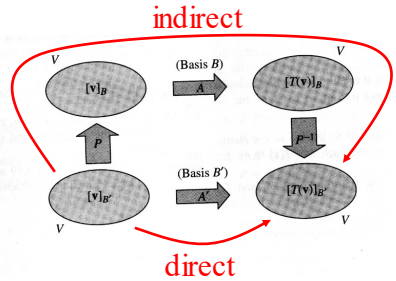
(1)(direct)

$A'[\mathbf{v}]_{B'} = [T(\mathbf{v})]_{B'}$

(2)(indirect)

$P^{-1}AP[\mathbf{v}]_{B'} = [T(\mathbf{v})]_{B'}$

$\Rightarrow \boxed{A' = P^{-1}AP}$



- Ex 1: (Finding a matrix for a linear transformation)

Find the matrix A' for $T: R^2 \to R^2$

$$T(x_1, x_2) = (2x_1 - 2x_2, -x_1 + 3x_2)$$

reletive to the basis $B' = \{(1,0), (1,1)\}$

Sol:

(I) $A' = [[T(1,0)]_{B'} \quad [T(1,1)]_{B'}]$

$$T(1,0) = (2,-1) = 3(1,0) - 1(1,1) \Rightarrow [T(1,0)]_{B'} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$T(1,1) = (0,2) = -2(1,0) + 2(1,1) \Rightarrow [T(1,1)]_{B'} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

$$\Rightarrow A' = [[T(1,0)]_{B'} \quad [T(1,1)]_{B'}] = \begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix}$$

(II) standard matrix for T (matrix of T relative to $B = \{(1,0), (0,1)\}$)

$$A = [T(1,0) \quad T(0,1)] = \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix}$$

transition matrix from B' to B

$$P = [[(1,0)]_B \quad [(1,1)]_B] = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

transition matrix from B to B'

$$P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

matrix of T relative B'

$$A' = P^{-1}AP = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix}$$

- Ex 2: (Finding a matrix for a linear transformation)

Let $B=\{(-3,2),(4,-2)\}$ and $B'=\{(-1,2),(2,-2)\}$ be basis for R^2, and let $A=\begin{bmatrix}-2 & 7\\ -3 & 7\end{bmatrix}$ be the matrix for $T: R^2 \to R^2$ relative to B.

Find the matrix of T relative to B'.

Sol:

transition matrix from B' to B: $P=\left[[(-1,2)]_B \quad [(2,-2)]_B\right]=\begin{bmatrix}3 & -2\\ 2 & -1\end{bmatrix}$

transition matrix from B to B': $P^{-1}=\left[[(-3,2)]_{B'} \quad [(4,-2)]_{B'}\right]=\begin{bmatrix}-1 & 2\\ -2 & 3\end{bmatrix}$

matrix of T relative to B':

$$A'=P^{-1}AP=\begin{bmatrix}-1 & 2\\ -2 & 3\end{bmatrix}\begin{bmatrix}-2 & 7\\ -3 & 7\end{bmatrix}\begin{bmatrix}3 & -2\\ 2 & -1\end{bmatrix}=\begin{bmatrix}2 & 1\\ -1 & 3\end{bmatrix}$$

- Ex 3: (Finding a matrix for a linear transformation)

For the linear transformation $T: R^2 \to R^2$ given in Ex.2, find $[\mathbf{v}]_B$, $[T(\mathbf{v})]_B$ and $[T(\mathbf{v})]_{B'}$, for the vector $\mathbf{v}$ whose coordinate matrix is

$$[\mathbf{v}]_{B'}=\begin{bmatrix}-3\\ -1\end{bmatrix}$$

Sol:

$$[\mathbf{v}]_B=P[\mathbf{v}]_{B'}=\begin{bmatrix}3 & -2\\ 2 & -1\end{bmatrix}\begin{bmatrix}-3\\ -1\end{bmatrix}=\begin{bmatrix}-7\\ -5\end{bmatrix}$$

$$[T(\mathbf{v})]_B=A[\mathbf{v}]_B=\begin{bmatrix}-2 & 7\\ -3 & 7\end{bmatrix}\begin{bmatrix}-7\\ -5\end{bmatrix}=\begin{bmatrix}-21\\ -14\end{bmatrix}$$

$$[T(\mathbf{v})]_{B'}=P^{-1}[T(\mathbf{v})]_B=\begin{bmatrix}-1 & 2\\ -2 & 3\end{bmatrix}\begin{bmatrix}-21\\ -14\end{bmatrix}=\begin{bmatrix}-7\\ 0\end{bmatrix}$$

or $[T(\mathbf{v})]_{B'}=A'[\mathbf{v}]_{B'}=\begin{bmatrix}2 & 1\\ -1 & 3\end{bmatrix}\begin{bmatrix}-3\\ -1\end{bmatrix}=\begin{bmatrix}-7\\ 0\end{bmatrix}$

- Similar matrix:

  For square matrices A and A' of order n, A' is said to be similar to A if there exist an invertible matrix P s.t. $A' = P^{-1}AP$

- Thm 6.13: (Properties of similar matrices)

  Let A, B, and C be square matrices of order n.
  Then the following properties are true.
  (1) A is similar to A.
  (2) If A is similar to B, then B is similar to A.
  (3) If A is similar to B and B is similar to C, then A is similar to C.

Pf:

$(1) A = I_n A I_n$

$(2) A = P^{-1}BP \Rightarrow PAP^{-1} = P(P^{-1}BP)P^{-1}$

$PAP^{-1} = B \Rightarrow Q^{-1}AQ = B \ (Q = P^{-1})$

- Ex 4: (Similar matrices)

$(a) A = \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix}$ and $A' = \begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix}$ are similar

because $A' = P^{-1}AP$, where $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$(b) A = \begin{bmatrix} -2 & 7 \\ -3 & 7 \end{bmatrix}$ and $A' = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ are similar

because $A' = P^{-1}AP$, where $P = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}$

- Ex 5: (A comparison of two matrices for a linear transformation)

Suppose $A=\begin{bmatrix}1&3&0\\3&1&0\\0&0&-2\end{bmatrix}$ is the matrix for $T: R^3 \to R^3$ relative to the standard basis. Find the matrix for T relative to the basis

$$B'=\{(1,1,0),(1,-1,0),(0,0,1)\}$$

Sol:

The transition matrix from B' to the standard matrix

$$P=\left[[(1,1,0)]_B \quad [(1,-1,0)]_B \quad [(0,0,1)]_B\right]=\begin{bmatrix}1&1&0\\1&-1&0\\0&0&1\end{bmatrix}$$

$$\Rightarrow P^{-1}=\begin{bmatrix}\frac{1}{2}&\frac{1}{2}&0\\\frac{1}{2}&-\frac{1}{2}&0\\0&0&1\end{bmatrix}$$

matrix of T relative to B':

$$A'=P^{-1}AP=\begin{bmatrix}\frac{1}{2}&\frac{1}{2}&0\\\frac{1}{2}&-\frac{1}{2}&0\\0&0&1\end{bmatrix}\begin{bmatrix}1&3&0\\3&1&0\\0&0&-2\end{bmatrix}\begin{bmatrix}1&1&0\\1&-1&0\\0&0&1\end{bmatrix}$$

$$=\begin{bmatrix}4&0&0\\0&-2&0\\0&0&-2\end{bmatrix}$$

- Notes: Computational advantages of diagonal matrices:

$$(1) D^k = \begin{bmatrix} d_1^k & 0 & \cdots & 0 \\ 0 & d_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n^k \end{bmatrix} \qquad D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}$$

$$(2) D^T = D$$

$$(3) D^{-1} = \begin{bmatrix} \frac{1}{d_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{d_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{d_n} \end{bmatrix}, \quad d_i \neq 0$$